

1. Selected Problem Statement

P3: Fraud & Double-Selling Detection

To eliminate land title fraud, illegal resale, and record contradictions in Bangladesh by implementing an automated Ownership Chain Contradiction Detection Engine capable of analyzing historical custody chains and ongoing transaction proposals.

2. Bangladesh Context

Land governance in Bangladesh is divided between two disparate governmental institutions: the *Directorate of Registration* (responsible for deed registration under the Ministry of Law) and the *Upazila Land Offices / Assistant Commissioner (Land)* (responsible for survey records and mutation/namjari under the Ministry of Land). Because these archives traditionally lacked real-time cross-verification, opportunistic syndicates exploit procedural lag to register unauthorized deeds and initiate conflicting mutations.

3. Problem Analysis

- **Double Selling:** Predatory resale of already transferred land to secondary buyers before the primary buyer's namjari is completed.
- **Ownership Chain Breaks:** Execution of transfer deeds by fictitious or unverified grantors lacking legitimate predecessor titles.
- **Area Inflation & Mismatches:** Transfer deeds stipulating acreage exceeding the parent cadastral survey khatian (CS, SA, RS, BS/City Survey).
- **Deed-Mutation Discrepancies:** Mutations granted for acreage or applicants inconsistent with registered deeds.
- **Judicial Burden:** Over 70% of pending civil litigation cases in Bangladesh courts originate directly from conflicting land records.

4. Proposed Solution: LandTrace

LandTrace is an automated Decision-Support System (DSS) tailored for Sub-Registrars and Assistant Commissioners (Land). It models land holdings as directed title graphs, verifying mathematical, spatial, and chronological integrity across five core fraud vectors in real time.

Land Records (Deeds & Khatians) → Data Normalization → Directed Ownership Chain Engine → Five-Vector Contradiction Audit → Risk Scoring (0-100) → Officer Decision Support

5. System Architecture

LandTrace operates as an integrated microservices-ready web platform deployed on scalable serverless cloud infrastructure:

- **Presentation Layer:** Bilingual administrative portal (বাংলা default / English) built with Next.js 16 App Router and Tailwind CSS.
- **Application & API Layer:** Server Components and route handlers executing the topological graph verification engine.
- **Data Persistence Layer:** Neon Serverless PostgreSQL with SSL pooling, managed by Prisma 6 ORM.
- **Continuous Deployment:** Git-driven automated pipeline via GitHub and Vercel.

6. Technology Stack

Layer	Technologies	Role in Architecture
Frontend	Next.js 16, React 19, Tailwind CSS v4	Bilingual officer portal, interactive timelines, print dossiers
Logic & Engine	TypeScript 5, Node Test Runner	Pure functional graph analysis and contradiction algorithms
Database	PostgreSQL 16 on Neon, Prisma ORM	Structured cadastral parcels, deeds, mutations, and audit logs
Hosting & CI/CD	Vercel, GitHub Actions	Automated zero-downtime production deployment

7. Key System Features

- **Bilingual Officer Portal:** Default বাংলা language designed to align with government terminologies (খতিয়ান, দাগ, মৌজা, নামজারি).
- **Interactive Title Timeline:** Visual chronological map of all transfers from initial grant to current claimant.
- **Pre-Registration Risk Check:** Decision-support audit tool allowing Sub-Registrars to test proposed transfers before execution.
- **Forensic Evidentiary Dossier:** Detailed discrepancy breakdown including affected parties, deed numbers, and exact decimal variances.
- **Immutable Officer Action Trail:** Action logging for reviewing records, requesting field inquiries, and adding notes.

8. Five Core Detection Capabilities

Capability	Detection Logic & Mathematical Trigger	Administrative Impact
1. Double Selling (দ্বৈত বিক্রয়)	Flags instances where Seller S transfers land to Buyer A, and subsequently executes deed to Buyer B without documented re-acquisition: $\text{Date}(\text{Sale } B) > \text{Date}(\text{Sale } A) \wedge \text{Title}(S) == \emptyset$	Prevents unauthorized secondary conveyances of sold parcels.
2. Chain Break (ধারাবাহিকতা বিচ্ছিন্ন)	Flags deeds where Grantor G in deed N does not appear as a Grantee in deed N-1, parent khatian, or inheritance record: $G \notin \text{ValidGranteeSet}(N-1)$	Stops fraudulent conveyances executed by phantom or impersonated grantors.
3. Area Mismatch (পরিমাপের গরমিল)	Flags conveyances where deed acreage exceeds the parent cadastral khatian area or cumulative active shares: $\text{Area}(\text{Deed}) > \text{Area}(\text{Khatian})$	Eliminates artificial acreage inflation and boundary encroachments.
4. Deed–Mutation Mismatch (দলিল-নামজারি অমিল)	Cross-validates AC-Land namjari records against sub-registry deeds for area discrepancies $(\text{Area}(\text{Mut}) - \text{Area}(\text{Deed}) > 0.05)$ or missing deeds.	Ensures mutated records strictly conform to legitimate registered deeds.
5. Wrong Timing / Rapid Flip (সময়ানুক্রমিক অসঙ্গতি)	Detects impossible chronology $(\text{Date}(\text{Sale}) < \text{Date}(\text{Acquisition}))$ or rapid speculative flipping $(\text{Interval} < 7 \text{ days})$ without interim namjari.	Flags backdated fraudulent deeds and rapid speculative syndicate activity.

9. Synthetic Data & Ethical Compliance

In accordance with hackathon guidelines, LandTrace operates strictly on **synthetic demonstration data**:

- **Zero Real Citizen Data:** All names, khatian numbers, and plot coordinates are fictional test cases.
- **Ethical Boundary:** No live government portal is scraped, and no production state database is accessed.
- **Benchmark Coverage:** Seven deterministic test parcels demonstrate valid chains and every permutation of fraud.

10. Countrywide Implementation Roadmap

Phase 1 (Months 1–4): Pilot in 2 Upazilas (Tejgaon & Sylhet Sadar) in parallel with e-Mutation. Phase 2 (Months 5–10): Divisional Integration with Land Commissionerate APIs and officer training. Phase 3 (Months 11–18): Nationwide rollout across all 64 districts with centralized Ministry oversight.

11. Legal & Regulatory Boundary

Decision-Support Notice: LandTrace is designed as an analytical decision-support tool for land officers. It does not issue judicial decrees, adjudicate legal ownership, or replace the statutory authority of Land Tribunals and Assistant Commissioners (Land).

12. Future Government Systems Integration Plan

[e-Registration (Sub-Registry)] — [Authorized API Integration] → [LandTrace Engine]
[e-Mutation (AC Land / Khatian)] — [Officer Review & Audit]

Note: Illustrates architectural target upon official government integration.

13. Team & Contact Information

Project	LandTrace
Problem Category	P3 — Fraud & Double-Selling Detection
Hackathon	LandTech Hackathon 2026
Live Production URL	https://landtrace-rafiul-kabir.vercel.app
GitHub Repository	https://github.com/RafiulKabir0/landtrace
Team	Team LandTrace